

# Bardia Amiryavari

Biomedical Engineer & Software Developer

Bern, Switzerland | Bardiaamiryavari@gmail.com | [linkedin.com/in/bardia-amiryavari](https://www.linkedin.com/in/bardia-amiryavari)

## WORK EXPERIENCE

2023 – Present

### Software Developer – Embedded Systems & IoT

*SensoPro GmbH, Switzerland*

- ROS2 (Python) · IMU sensor integration · real-time gait analysis · Raspberry Pi
- Flask REST APIs · remote device management · network monitoring
- React TypeScript dashboards · sensor data visualization · KPI metrics
- Docker on ARM · OTA deployment via Mender · CI/CD pipelines
- Tailscale VPN · SSH · headless Ubuntu · Azure DevOps · Git · MkDocs

2024 – 2025

### Research Assistant

*Berner Fachhochschule rehaLab, Biel, Switzerland*

- Rehabilitation robotics · assistive technologies · movement analysis

2018 – 2021

### Human Resources Coordinator – Volunteers

*Mahak Charity (International Society for Children with Cancer), Tehran, Iran*

- Volunteer recruitment · scheduling · team coordination · NGO operations

2013 – 2021

### Factory Manager

*Nanetehran Confectionary & Bakery, Tehran, Iran — Promoted through 6 roles over 8 years*

- Factory Manager (Apr 2017 – Sep 2021) · Production planning · factory operations · multi-site management
- Store Manager (Oct 2017 – Apr 2019) · Store management · planning · team leadership
- Accounting Assistant (Apr 2016 – Oct 2017) · Budgeting · financial records
- Assistant Bakery Manager (Dec 2014 – Apr 2016) · HR · production support · baking operations
- Assistant Store Manager (Mar 2014 – Dec 2014) · Store & order management
- Store Cashier (Oct 2013 – Mar 2014) · Customer service · cash register

## EDUCATION

Sep. 2021 – 2024

### M.Sc. Biomedical Engineering – Biomechanics

*University of Bern / Bern University of Applied Sciences (BFH), Switzerland*

Thesis: Vision-based Pose Estimation Platform for Rehabilitation Applications

Supervisors: Prof. Dr. Juan Fang, Dr. Efe Anil Aksöz — Institute for Human Centered Engineering, BFH

Funded by Innosuisse (application no. 102.229 IP-LS)

- Developed a real-time 3D joint tracking system using MediaPipe, MMPose, VitPose, RTMPose, and MotionBert, combined with Intel RealSense depth cameras and Python.
- Validated against Vicon marker-based motion capture and IMUs; achieved joint angle RMSE within 10 degrees and Pearson Correlation Coefficient consistently above 0.9.
- Integrated the platform with the SensoPro and Juandago rehabilitation device prototypes, and developed a bio-feedback UI with live visual feedback, performance tracking, and an interactive therapy game.

Courses: Movement Biomechanics · Solid Mechanics · Intro to AI · Finite Element · Rehabilitation Technology

Sep. 2014 – Sep.  
2018

## **B.Sc. Biomedical Engineering – Biomechanics**

*Azad University, Science and Research Branch, Tehran, Iran*

Thesis: Spinal Thoracic Bone Recovery in Scoliosis Patients — Pre- and Post-operative Analysis

Courses: Prosthetics & Orthotics · Intro Biomechanics · Numerical Methods

2014

## **Diploma – Mathematics and Physics**

*Mesbah Educational Complex, Tehran, Iran*

## TECHNICAL SKILLS

|                         |                                                                                |
|-------------------------|--------------------------------------------------------------------------------|
| <b>Programming</b>      | Python · JavaScript / TypeScript · MATLAB · TwinCat · Bash                     |
| <b>Frameworks</b>       | ROS2 (rclpy) · Flask · React · MkDocs                                          |
| <b>DevOps &amp; IoT</b> | Docker · Git · Azure DevOps · Mender (OTA) · Tailscale · SSH · CI/CD Pipelines |
| <b>Hardware</b>         | Raspberry Pi · IMU Sensors · Embedded Linux (Ubuntu) · USB/Ethernet Interfaces |
| <b>CAD / FEA</b>        | SolidWorks · Shapr3D · Abaqus                                                  |
| <b>Office</b>           | Microsoft Office Suite                                                         |

## LANGUAGES

|                |                   |
|----------------|-------------------|
| <b>Persian</b> | Native            |
| <b>English</b> | Fluent (C1)       |
| <b>French</b>  | Advanced (B2)     |
| <b>German</b>  | Intermediate (B1) |

## EVENTS & ADDITIONAL TRAINING

|      |                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------|
| 2022 | <b>Rehabilitation Robotics – FES and EMG for Rehabilitation</b><br><i>University of Bern</i>              |
| 2022 | <b>Practical Orthopedic Course – Surgery Room &amp; Patient Consultation</b><br><i>University of Bern</i> |
| 2012 | <b>Austrian Young Physics Tournament (AYPT)</b><br><i>Vienna, Austria</i>                                 |
| 2024 | <b>Foundations of User Experience (UX) Design</b><br><i>Google · Coursera</i>                             |

## INTERESTS

Photography · Martial Arts · Drums